

2024, Mar. 2024 Release LOC:Acute Pediatric**Diabetic Ketoacidosis****Utilization Benchmarks****Length of Stay:** 2.2 d**Length of Stay Type:** InterQual**Condition/Procedure:** DIABETIC KETOACIDOSIS**% Paid as Observation:** 12%

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Overview**Select Day****Episode Day 1****Episode Day 2****Episode Day 3****Discharge Screens** ⁽⁵⁰⁾**Notes**

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Overview**Introduction:**

Diabetic ketoacidosis (DKA) is a serious acute metabolic complication of diabetes mellitus characterized by the metabolic triad of hyperglycemia, anion gap metabolic acidosis, and ketosis. DKA occurs when there is a relative or absolute decrease in insulin levels and an excess of counterregulatory hormones. As a result of this imbalance, gluconeogenesis and glycogenolysis occur with increased glucose production and decreased peripheral glucose utilization. This subsequently leads to hyperglycemia, hyperosmolality, increased lipolysis, and ketogenesis, thereby causing ketosis and metabolic acidosis. Despite improvements in diabetes care, DKA remains a life-threatening condition that, in most cases, requires urgent hospitalization for monitoring and management to correct dehydration, ketosis, and acidosis (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Glaser et al., Pediatr Diabetes 2022, 23: 835-56; Cashen and Petersen, Pediatrics in review 2019, 40: 412-20).

Evaluation and Treatment:

Approximately 15-70% of individuals with type 1 diabetes mellitus (DM) may develop diabetic ketoacidosis (DKA). Groups experiencing disproportionate prevalence of DKA include children of certain ethnic groups and those young in age. Patients with type 2 DM may also present with DKA. The frequency of type 2 diabetes in the pediatric population is increasing worldwide. Overall, 5% to 25% of children with type 2 diabetes have DKA at the time of diagnosis. Precipitating factors may include new-onset diabetes, inadequate management of infection, insufficient insulin administration, malfunctioning insulin pump, pregnancy, and specific medication reactions (i.e., high-dose corticosteroids and immunosuppressive medications). There is substantial variability in the presentation of DKA that may not include the classic symptoms associated with hyperglycemia. Therefore, individualization of treatment is imperative. Clinical manifestations of DKA may include mental status changes and neurological abnormalities, dehydration, nausea, vomiting, abdominal pain, and Kussmaul respirations. The diagnosis of DKA is based on the triad of laboratory findings of elevated glucose, ketonemia or ketonuria, and acidosis. According to the International Society for Pediatric and Adolescent Diabetes, the biochemical definition of DKA includes the following:

- A blood glucose level > 200 mg/dl (11 mmol/L)

- Venous pH < 7.3 or serum bicarbonate < 18 mEq/L (18 mmol/L)
- Ketonemia or moderate or large ketonuria

According to the International Society for Pediatric and Adolescent Diabetes, the severity of DKA is defined as:

- Mild: venous pH < 7.3 or serum bicarbonate < 18 mEq/L (18 mmol/L)
- Moderate: venous pH < 7.2 or serum bicarbonate < 10 mEq/L (10 mmol/L)
- Severe: venous pH < 7.1 or serum bicarbonate < 5 mEq/L (5 mmol/L)

While most patients with DKA have significantly elevated glucose values, around 10% of patients may present with euglycemic diabetic ketoacidosis in which they have a lower blood glucose level than anticipated. The etiology of euglycemic ketoacidosis includes but is not limited to pregnancy, decreased caloric intake, alcohol use disorders, chronic liver disease, and insulin administration en route to the hospital. Successful treatment of DKA requires addressing the patient's acidosis, ketosis, hyperosmolality, and hyperglycemia which involves monitoring and correcting electrolyte abnormalities as needed, intravenous fluid resuscitation, insulin administration, monitoring for cerebral edema, and identification and treatment of precipitating events. DKA is associated with many complications. Cerebral injury is a complication that primarily contributes to morbidity and mortality in DKA, with a mortality rate of 21%-24%.

High-quality diabetes care requires addressing barriers impeding disease management and providing diabetes self-management education and support (DSMES) that is specific to the needs of each individual patient. The times of diagnosis and when acute complications arise represent critical moments to evaluate knowledge deficits and needs and intervene with DSMES that should begin immediately and continue throughout the patient's journey across the care continuum. Education provided in a comprehensive and easy-to-understand format without unnecessary complexity can prevent the recurrence of DKA and other complications and improve patient self-management, health status, and clinical outcomes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Glaser et al., Pediatr Diabetes 2022, 23: 835-56; Cashen and Petersen, Pediatrics in review 2019, 40: 412-20; Dhatariya and Vellanki, Curr Diab Rep 2017, 17: 33).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content in this subset is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Episode Day 1, One:**

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽¹⁾

- Blood glucose > 200 mg/dL(11 mmol/L) ⁽²⁾
- Ketones, elevated ^(3, 4, 5)

● **Acidosis ≥ One:** ⁽⁶⁾

- Anion gap 16-18 mEq/L(16-18 mmol/L) ^(7, 8)
- Arterial or venous pH 7.2-7.29 ⁽⁹⁾
- Serum HCO₃ or CO₂ 10-17 mEq/L(10-17 mmol/L) ⁽¹⁰⁾

● **Intervention, Both:** ⁽¹¹⁾

- Insulin (SC only) at least every 4h ^(12, 13)
- IV fluid resuscitation ^(14, 15)

● **CRITICAL, Both:** ⁽¹⁶⁾● **Finding, ≥ One:** ⁽¹⁷⁾● **Labs, All:**

- Blood glucose > 200 mg/dL(11 mmol/L)
- Ketones, elevated ^(3, 4, 5)

● **Acidosis ≥ One:** ⁽⁶⁾

- Anion gap > 18 mEq/L(18 mmol/L) ^(7, 8)
- Arterial or venous pH < 7.2 ⁽⁹⁾
- Serum HCO₃ or CO₂ < 10 mEq/L(10 mmol/L) ⁽¹⁰⁾

- Age < 5 years

- Coma, stupor, or somnolence ⁽¹⁸⁾- Pregnant individual ⁽¹⁹⁾● **Intervention, Both:** ⁽¹¹⁾

- Insulin, continuous ^(12, 20, 21)
- IV fluid resuscitation ^(14, 15)

● **Episode Day 2, One:**● **OBSERVATION, One:**● **Responder**, discharge expected today if clinically stable for 12h, **All:** ⁽²²⁾

- Blood glucose within acceptable limits ⁽²³⁾
- Lab values within acceptable limits ^(23, 24)
- Tolerating PO ⁽²⁵⁾

● **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**● **Blood glucose, All:**● **Finding, ≥ One:**

- > 250 mg/dL(13.9 mmol/L)
- < 70 mg/dL(3.9 mmol/L) ⁽²⁶⁾

- Blood glucose monitoring ^(27, 28)- Insulin or oral glucose-lowering medication initiation, adjustment, or dose held ⁽²⁹⁾● **Hypokalemia, Both:** ^(30, 31, 32)

- Potassium 2.5-3.2 mEq/L(2.5-3.2 mmol/L)
- Potassium repletion (includes PO)

● **Management of precipitating cause of diabetic ketoacidosis, ≥ One:** ⁽³³⁾

- Infection and anti-infective ≤ 24h
- Pump failure actual or suspected and transition to insulin (SC only) or back to insulin pump ≤ 24h
- Patient or caregiver knowledge deficit and diabetes education ^(34, 35)

● **ACUTE, One:** ⁽³⁶⁾

- Post Critical care ≤ 24h

● **INTERMEDIATE, ≥ One:** ⁽³⁷⁾● **HFNC and, ≥ One:** ^(38, 39)

- Flow rate, **One:** ⁽⁴⁰⁾
 - ≤ 20 L/min (> 10 kg)
 - < 2 L/kg/min (≤ 10 kg)
 - $\text{FiO}_2 > 21\%(0.21)$ to $39\%(0.39)$ ⁽⁴¹⁾
- NIPPV ⁽⁴²⁾
- Oxygen $35\text{--}39\%(0.35\text{--}0.39)$ ⁽⁴¹⁾
- **CRITICAL, $\geq$ One:**
 - Diabetic ketoacidosis and insulin, continuous ^(12, 20)
 - HFNC and, $\geq$ **One:** ^(38, 39)
 - Flow rate, **One:** ⁽⁴³⁾
 - > 20 L/min (> 10 kg)
 - ≥ 2 L/kg/min (≤ 10 kg)
 - $\text{FiO}_2 \geq 40\%(0.40)$ ⁽⁴¹⁾
 - Hypokalemia, **All:** ^(30, 31, 32)
 - Potassium < 3.2 mEq/L (3.2 mmol/L)
 - ECG findings, $\geq$ **One:** ⁽³²⁾
 - Multifocal PVCs
 - PVCs $> 6/\text{min}$
 - R on T phenomenon ⁽⁴⁴⁾
 - T and U wave fusion ⁽⁴⁵⁾
 - U wave ⁽⁴⁶⁾
 - Potassium repletion
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽⁴²⁾
 - $\text{FiO}_2 \geq 40\%(0.40)$
 - Respiratory intervention every 1-2h ⁽⁴⁷⁾
 - Oxygen $\geq 40\%(0.40)$ ⁽⁴¹⁾
- **Episode Day 3, One:**
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable for 12h, **All:** ⁽²²⁾
 - Blood glucose within acceptable limits ⁽²³⁾
 - Lab values within acceptable limits ^(23, 24)
 - Patient or caregiver knowledgeable about disease and understands the regimen ⁽⁴⁸⁾
 - Tolerating PO ⁽²⁵⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
 - For a condition other than Diabetic ketoacidosis use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 24h, **All:** ⁽²²⁾
 - Blood glucose within acceptable limits ⁽²³⁾
 - Lab values within acceptable limits ^(23, 24)
 - Tolerating PO ⁽²⁵⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁴⁹⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **INTERMEDIATE, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
 - **Use Extended Stay subset**

- Refer for secondary review

● **CRITICAL, One:**

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
-

Discharge, Level of Care, One: ⁽⁵⁰⁾

● **HOME, Both:**

- Level of care appropriateness, **Both:**
 - Home environment safe and accessible ⁽⁵¹⁾
 - Patient or caregiver demonstrates ability to manage care
- Diabetic ketoacidosis (DKA) discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁵²⁾
 - Education on the importance of, **All:**
 - Prevention and treatment of hypoglycemia
 - Home monitoring of blood glucose
 - Sick day management
 - Foot inspection and care
 - Annual influenza vaccine
 - Annual eye examination
 - Lifestyle modification to prevent hypertension and dyslipidemia ⁽⁵³⁾
 - Smoking cessation, if indicated
 - Glycemic control and hypertension management, if hypertension present, to prevent retinopathy, nephropathy, and neuropathy ⁽⁵⁴⁾
 - Medication reconciliation ⁽⁵⁵⁾
 - Follow-up care planned ⁽⁵⁶⁾
 - Identify and address transportation needs

● **HOME CARE, Both:**

- Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽⁵¹⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(57, 58)
- Skilled services, ≥ **One:** ⁽⁵⁹⁾
 - Skilled nursing ⁽⁶⁰⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁶¹⁾
- Diabetic ketoacidosis (DKA) discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁵²⁾
 - Education on the importance of, **All:**
 - Prevention and treatment of hypoglycemia
 - Home monitoring of blood glucose
 - Sick day management
 - Foot inspection and care
 - Annual influenza vaccine
 - Annual eye examination
 - Lifestyle modification to prevent hypertension and dyslipidemia ⁽⁵³⁾
 - Smoking cessation, if indicated
 - Glycemic control and hypertension management, if hypertension present, to prevent retinopathy, nephropathy, and neuropathy ⁽⁵⁴⁾
 - Medication reconciliation ⁽⁵⁵⁾
 - Follow-up care planned ⁽⁵⁶⁾
 - Identify and address transportation needs

● BEHAVIORAL HEALTH, Both:

- Level of care appropriateness, **One:**
 - Support systems available ⁽⁶²⁾
- Skilled treatment, **≥ One:**
 - Clinical assessment ⁽⁶³⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁶⁴⁾

● SKILLED MEDICAL OR THERAPY, Both:

- Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 1\text{x/wk}$
 - Treatment precluded in a lower level
- Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment $1\text{-}2\text{x}/24\text{h}$ ⁽⁶⁰⁾
- Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - Functional impairments requiring at least supervision ⁽⁶⁷⁾
 - Able to tolerate 1h/d of skilled therapy $\geq 5\text{d/wk}$
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁵⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

● SUBACUTE MEDICAL OR THERAPY, Both:

- Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 2\text{x/wk}$
 - Treatment precluded in a lower level
- Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment $4\text{h}/24\text{h}$ ⁽⁶⁰⁾
- Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾
 - Able to tolerate $2\text{-}3\text{h/d}$ of skilled therapy $\geq 5\text{d/wk}$ and ≥ 1 therapy discipline
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁵⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

● SUBACUTE REHAB, Both:

- Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$
 - Treatment precluded in a lower level
- Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾
 - Able to tolerate $2\text{-}3\text{h/d}$ of skilled therapy $\geq 5\text{d/wk}$ and ≥ 2 therapy disciplines
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁵⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

● ACUTE REHAB, Both:

- Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$

- **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾
 - Able to tolerate ≥ 15 h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁶⁸⁾
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁵⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level
 - **In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁶⁹⁾
 - Ventilator dependent ≥ 6 h/d and weaning planned ^(70, 71)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁷²⁾
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁵⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Notes:**1:**

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

2:

The biochemical definition of diabetic ketoacidosis (DKA) includes a blood glucose level greater than 200 mg/dL (11 mmol/L), venous pH less than 7.3 or serum bicarbonate less than 18 mEq/L (18 mmol/L), with ketonemia or ketonuria. Mild DKA findings include a venous pH of 7.2 to 7.29 or serum bicarbonate of 10 mEq/L (10 mmol/L) to 17 mEq/L (17 mmol/L) (Glaser et al., *Pediatr Diabetes* 2022, 23: 835-56).

3:

Diagnostic testing for elevated ketone levels includes evaluation of one or more of the following:

- Beta-hydroxybutyrate > 3 mmol/L (31 mg/dL)
- Serum ketones positive (ketonemia)
- Urine ketones positive (ketonuria) ≥ 2+ (moderate to large)

4:

Urine ketone testing detects the presence of acetoacetate and acetone. In diabetic ketoacidosis, positive urine ketones findings are usually ≥ 2+ (moderate or large) (Glaser et al., *Pediatr Diabetes* 2022, 23: 835-56).

5:

A blood beta-hydroxybutyrate level greater than or equal to 3 mmol/L is indicative of DKA (Kilpatrick et al., *Diabetes Care* 2022, 45: 267-72).

6:

In diabetic ketoacidosis the combination of insulin deficiency and higher counterregulatory hormone concentrations increase lipolysis and ketogenesis resulting in ketonemia and metabolic acidosis (Glaser et al., *Pediatr Diabetes* 2022, 23: 835-56).

7:

The anion gap is one of the important biochemical parameters of diabetic ketoacidosis reflecting ketonemia and can be used to evaluate and determine the resolution of ketoacidosis (Glaser et al., *Pediatr Diabetes* 2022, 23: 835-56; Cashen and Petersen, *Pediatrics* in review 2019, 40: 412-20; von Oettingen et al., *Diabetes research and clinical practice* 2018, 135: 76-84).

8:

A normal anion gap is 12 mEq/L (12 mmol/L), plus or minus, 2 mEq/L (2 mmol/L). The anion gap is calculated using the following formula: $\text{anion gap} = \text{Na} - (\text{Cl} + \text{HCO}_3)$ (Glaser et al., *Pediatr Diabetes* 2022, 23: 835-56; Cashen and Petersen, *Pediatrics* in review 2019, 40: 412-20).

9:

Venous blood gases (VBG) may be used to assess a patient's ventilation or metabolic status. Venous pH or HCO_3 values closely approximate corresponding values derived from arterial blood gases (ABG) in conditions such as chronic obstructive pulmonary disease (COPD), respiratory distress syndrome, neonatal sepsis, renal failure, pneumonia, diabetic ketoacidosis, and status epilepticus (McKeever et al., *Thorax* 2016, 71: 210-5).

10:

Measurement of bicarbonate (HCO_3^-) in venous blood is done indirectly via a technique that measures carbon dioxide (CO_2). Total CO_2 may be slightly higher, up to 5%(0.05) above the actual HCO_3^- , due to other substances in the plasma. Since the difference between total CO_2 and actual HCO_3^- is small, the 2 quantities are usually treated as being equal.

11:

Diabetic ketoacidosis (DKA) treatment and management goals include restoring circulatory volume and tissue perfusion, correcting hyperglycemia, electrolyte imbalances, and acidosis, and addressing the underlying cause of DKA (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

12:

Insulin therapy is essential to diabetic ketoacidosis treatment to return cellular metabolism to normal, normalize blood glucose levels, and suppress lipolysis and ketogenesis (Glaser et al., Pediatr Diabetes 2022, 23: 835-56).

13:

Evidence has shown that subcutaneous (SC) insulin treatment for mild diabetic ketoacidosis (DKA) is safe. However, SC insulin treatment for mild DKA is contraindicated in children with impaired peripheral circulation. It is essential to provide adequate fluid replacement, frequent blood glucose monitoring, management of concurrent illness and precipitating factors, and appropriate, timely follow-up with the use of SC insulin treatment for DKA management (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Glaser et al., Pediatr Diabetes 2022, 23: 835-56).

14:

Administration of intravenous fluid is the first line of treatment for diabetic ketoacidosis (DKA) to restore intravascular volume, lower blood glucose, ensure adequate tissue perfusion, and assist in resolving metabolic acidosis. A 0.9% sodium chloride solution is recommended for initial fluid resuscitation. For children in DKA with volume depletion without shock, an initial fluid bolus of 10 to 20 mL/kg infused over 20-30 minutes is recommended to restore peripheral circulation. An initial bolus volume of 20 mL/kg is recommended if tissue perfusion is poor. After fluid resuscitation, fluid deficit replacement can be achieved with 0.45%-0.9% saline or a balanced salt solution. The goal is to correct estimated fluid deficits within 48 hours. As hyperglycemia improves, intravenous fluids with dextrose should be administered simultaneously with insulin until DKA resolves to avoid hypoglycemia (Glaser et al., Pediatr Diabetes 2022, 23: 835-56).

15:

For children in diabetic ketoacidosis with the rare instance of shock, 0.9% saline administered in 20 mL/kg boluses infused as fast as possible through a large bore cannula with a reassessment of circulatory status after every bolus is recommended for rapid restoration of circulatory volume (Glaser et al., Pediatr Diabetes 2022, 23: 835-56).

16:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

17:

The biochemical definition of diabetic ketoacidosis (DKA) includes a blood glucose level greater than 200 mg/dL (11 mmol/L), venous pH less than 7.3 or serum bicarbonate less than 18 mEq/L (18 mmol/L), with ketonemia or ketonuria. The level of acidosis defines the severity of DKA. Moderate DKA findings include pH less than 7.2 or serum bicarbonate less than 10 mEq/L (10 mmol/L). Severe DKA findings include pH less than 7.1 or serum bicarbonate less than 5 mEq/L (5 mmol/L). Children with severe DKA or those at increased risk for complications (e.g., age less than 5 years, pregnant individual) are appropriate for treatment and management at the Critical level of care (Glaser et al., Pediatr Diabetes 2022, 23: 835-56).

Note: InterQual® external peer reviewers confirm that in addition to severe DKA and those with increased risk for

complications, children with moderate DKA are also appropriate for management at the Critical level of care.

18:

Acute mental status change may include coma, stupor, or somnolence (Keith Kleinman, In: Harriet Lane Handbook, 22nd edn. 2021).

Instruction: Patients with acute coma, stupor, or somnolence should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma or stupor.

19:

Pregnancy is a ketogenic state that increases the risk of diabetic ketoacidosis (DKA) for individuals with a history of type 1 or type 2 diabetes mellitus who become pregnant. DKA can manifest in a euglycemic state for pregnant individuals and potentially be unrecognized. DKA increases the risk of stillbirth and puts the mother and fetus at a higher risk of morbidity and mortality (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

20:

Intravenous insulin should be administered to patients with diabetic ketoacidosis (DKA) until resolution. Resolution of DKA includes (Glaser et al., Pediatr Diabetes 2022, 23: 835-56):

- Serum bicarbonate level > 18 mEq/L (18 mmol/L)
- Venous pH > 7.30
- Beta-hydroxybutyrate < 1 mg/dL (1 mmol/L) or closure of the anion gap

Following resolution of DKA, subcutaneous insulin therapy can be initiated. When transitioning to SC insulin, a dose of basal (long-acting) insulin should be administered in addition to rapid or short-acting insulin. Alternatively, basal insulin can be given while the intravenous insulin is infusing. The first SC insulin may be given 15-30 minutes (with rapid-acting insulin) before the cessation of the insulin infusion to allow sufficient time for insulin absorption and prevention of rebound hyperglycemia. With long-acting insulin, the overlap should be longer with a gradual rate reduction of the insulin infusion. The regimen, dose, and type of SC insulin should be individualized. After transitioning to SC insulin, frequent glucose monitoring for hyperglycemia and hypoglycemia is required (Glaser et al., Pediatr Diabetes 2022, 23: 835-56).

21:

The administration of insulin via continuous intravenous infusion is the standard of care and the most effective method for achieving glycemic targets and closing the anion gap for critically ill individuals with diabetic ketoacidosis (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

22:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

23:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

24:

A glucose target between 100-180 mg/dL (5.6-10.0 mmol/L) is recommended for most patients in noncritical care units (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

25:

Oral intake should be greater than or equal to the patient's maintenance fluid requirements or within parameters which the medical practitioner determines are acceptable. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid requirements (mL/kg/d) can be calculated based on current body weight (kg):

- 100 mL/kg/d for first 10 kg
- 50 mL/kg/d for second 10 kg

- 20 mL/kg/d for each additional kg in weight

26:

The risk of hypoglycemia increases in young children (less than six years of age) who often cannot recognize or communicate their needs or manage hypoglycemia. There is a strong correlation between the frequency of blood glucose monitoring and glucose management. Therefore, confirmation of hypoglycemia may require frequent blood glucose monitoring in young children and infants (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Abraham et al., Pediatr Diabetes 2018, 19 Suppl 27: 178-92).

27:

The target glycemic range for non-critically ill hospitalized patients with diabetes is 100-180 mg/dL (5.6-10.0 mmol/L). Blood glucose concentrations greater than 180 mg/dL (10.0 mmol/L) may be acceptable, at times, to reduce the risk of harmful outcomes in patients with severe comorbidities or decreased life expectancy. Blood glucose targets should be individualized and modified if there are identifiable risks for adverse effects such as a history of frequent hypoglycemia or hypoglycemia unawareness (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

28:

In non-critically ill hospitalized patients with adequate nutritional intake, recommendations are to complete point-of-care (POC) glucose testing immediately before meals. Alternatively, in patients with inadequate oral intake, oral intake restrictions, or those on continuous enteral or parenteral nutrition, recommendations are to complete POC glucose testing every 4-6 hours. However, blood glucose monitoring is individualized. Additional or more frequent blood glucose testing is necessary if the patient cannot effectively communicate their needs, recognize signs or symptoms of a complication, or has a history of increased risk of diabetes-related adverse events (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

29:

Instruction: Medication adjustment includes initiation of a new drug, change in dosage, and/or administration frequency of a medication. Blood glucose values guide medication dosing decisions and adjustments. Administration of sliding scale insulin must occur at least 3 times within 24 hours. A continuous infusion rate must have at least 2 rate adjustments within 24 hours (including insulin pump rate adjustments). A glucose-lowering medication dose withheld based on normal or low blood glucose levels is included in the count of doses administered within the 24-hour timeframe.

30:

Hypokalemia rarely occurs in isolation and is usually secondary to a disease process or overuse of diuretics. Criteria should be reviewed for the underlying cause of the hypokalemia (e.g., diabetic ketoacidosis, gastroenteritis) to ensure the patient is managed appropriately.

31:

When a patient with hypokalemia is not responding to potassium repletion, the magnesium level should be checked since hypomagnesemia could be contributing to the refractory hypokalemia.

32:

Hypokalemia (Sandau et al., Circulation 2017, 136: e273-e344; Zieg et al., Acta Paediatr 2016, 105: 762-72)

33:

Infection and malfunction of diabetes technology (e.g., insulin pump), amongst others, are common precipitating causes of diabetic ketoacidosis (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; French et al., Bmj 2019, 365: l1114).

34:

National standards recommend diabetes self-management education and support (DSMES) for patients and/or caregivers to enhance knowledge, decision making, and mastery of skills for diabetes management to prevent acute

complications, improve treatment outcomes, and reduce risk of long-term complications. Hospitalization represents a critical time to evaluate, assess, and provide DSMES that includes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314):

- Identifying a health care provider to provide diabetes care post discharge
- Establishing home blood glucose goals, self-monitoring of blood glucose, and understanding when to call the provider
- Recognition, treatment, and prevention of hyperglycemia and hypoglycemia
- Administration of blood glucose-lowering medications, including insulin, if relevant
- Information on healthy dietary choices
- Proper use and disposal of insulin needles and syringes
- Sick-day management (e.g., when sick with a cold, flu, or infection)

35:

Critical events such as a new diagnosis, unmet treatment goals, complications, and transitions in life require evaluation of diabetes self-management educational needs to encourage acquiring the skills that may assist in executing and adhering to the treatment regimen and medical nutrition therapy (MNT) plan. An individualized MNT plan created through a joint effort of shared decision making with the patient and the healthcare team that includes a registered dietitian nutritionist is essential. The goals of MNT are to attain individualized glycemic targets, delay or prevent complications, reach and maintain body weight goals, incorporate personal and cultural preferences, address barriers to behavioral changes and access to healthy foods, and provide the tools and resources to develop and maintain healthy dietary patterns. Individualized macronutrient recommendations consider the patient's eating patterns, preferences, and metabolic goals, as evidence shows there are no general recommendations for macronutrients for all people with diabetes. Carbohydrate monitoring strategies are essential to achieve glycemic control and include, but are not limited to, carbohydrate counting, an exchange system, or experience-based estimation (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; American Diabetes Association, Diabetes Care 2022, 45:S1-S262).

36:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

37:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

38:

High flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates set independent from the FiO_2 . The heated and humidified gas has been shown to improve comfort, aid in the clearance of secretions, and reduce bronchospasms. When flow rates are set above the patient's inspiratory demand, the upper airway is washed out of carbon dioxide, nasal resistance is decreased, and there is a reduction in dead space, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. Bronchiolitis is the main indication for HFNC in pediatric patients beyond the neonatal period. Although the benefits of HFNC in other conditions such as pneumonia, asthma, and post-extubation have been reported, more well-designed studies are needed to guide the use of HFNC in these conditions (Kwon, Clin Exp Pediatr 2020, 63: 3-7).

39:

Currently, there is no universally accepted method for weaning pediatric patients off high-flow nasal cannula (HFNC), and weaning protocols vary based on institutional policies. However, one commonly used method is to gradually decrease the fraction of inspired oxygen (FiO_2) to a specific level, such as 40%(0.40), then gradually reduce the flow rate before transitioning to a low-flow oxygen delivery system (e.g., nasal cannula). Less common

methods include weaning the FiO_2 to a predetermined value, then transitioning to a low-flow oxygen system without reducing the flow rate, or gradually reducing the flow rate while keeping the FiO_2 constant (Kawaguchi et al., *Pediatr Crit Care Med* 2020, 21: e228-e35). There is no standard flow rate at which HFNC should be discontinued, but based on the literature, it is suggested to discontinue HFNC when the flow rate reaches 2 L/min in infants, and 4 L/min in children older than 12 months (Udurgucu et al., *Pediatr Allergy Immunol Pulmonol* 2022, 35: 79-85; Peterson et al., *Respir Care* 2021, 66: 591-9).

Instruction: When weaning a patient off high-flow nasal cannula (HFNC), it is expected that the current level of care be maintained or a lower level of care from the previous level be utilized to conduct the review if HFNC criteria are used. A typical weaning strategy usually does not involve simultaneously decreasing the flow rate while increasing the FiO_2 . This scenario may indicate that a patient is being kept on the HFNC machine but may be able to tolerate settings that could be administered by nasal cannula. For instance, a patient who was previously on HFNC 10 L/min, 30%(0.30) FiO_2 might be weaned to 3 L/min, but FiO_2 is increased to 100%(1.0). In this case, the patient should be assessed based on the flow rate rather than the FiO_2 .

40:

High flow nasal cannula (HFNC) consists of three parameters: temperature, fraction of inspired oxygen (FiO_2), and flow rate. Each parameter is independent of the other. The flow rate is how fast the heated and humidified air/oxygen is delivered per minute and is documented as liters per minute (L/min). Standard HFNC flow rates in pediatrics have yet to be established (Kwon, *Clin Exp Pediatr* 2020, 63: 3-7). Although most studies on HFNC in pediatrics focus on children with bronchiolitis, they can provide some guidance on appropriate flow rates outside of the intensive care unit (ICU). Initial flow rates in clinical trials have found that 1-2 L/kg/min is effective in patients with bronchiolitis (Yurtseven et al., *Pediatr Pulmonol* 2019, 54: 894-900). However, a recent survey revealed that only 19%(0.19) of responders use a weight-based flow rate method (Miller et al., *Respir Care* 2018, 63: 894-9). This method may not be optimal in older children as it can result in very high flow rates that may not be tolerated. Some studies combat this potential problem by setting maximum flow rates. Flow rates up to 20 L/min are well tolerated on non-ICU wards (Willer et al., *Hosp Pediatr* 2021, 11: 891-5; Kepreotes et al., *Lancet* 2017, 389: 930-9). Therefore, a flow rate of 20 L/min was chosen as the maximum for the Intermediate level of care.

Instruction: To calculate the L/kg/min, the flow rate is divided by the patient's weight.

Example:

HFNC flow rate: 12 L/min

Patient weight: 8 kg

$12/8 = 1.5$

This patient is on a flow rate of 1.5 L/kg/min

41:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air [room air: 21%(0.21) FiO_2] to correct hypoxia. For pediatric patients, the selection of an oxygen delivery system needs to take into consideration the patient's size, needs, and therapeutic goals. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, delivery device, patient's minute ventilation (dependent on the respiratory rate and tidal volume), and oxygen flow rate. In general, patients with higher minute ventilation (common in most sick patients and those with respiratory disease) require considerably higher flow rates to correct hypoxia. The following are examples of oxygen delivery systems commonly used for pediatric patients with the percentage range of FiO_2 optimally delivered:

Low-flow systems

- Nasal cannula / Nasopharyngeal catheters 24-35%(0.24-0.35) FiO_2

Reservoir systems (For neonates, infants, and young children, the maximum FiO_2 achievable is not well documented)

- Simple oxygen masks 35-50%(0.35-0.50) FiO_2
- Partial-rebreathing masks 40-60%(0.40-0.60) FiO_2
- Non-rebreathing masks > 60%(0.60) FiO_2

High-flow systems

- Air-entrainment masks / nebulizers 24-40%(0.24-0.40) FiO_2
- Heated, humidified, high flow oxygen concentrating nasal cannula 21-100%(0.21-1.0) FiO_2

Enclosure systems

- Oxygen hoods / tents 21-100%(0.21-1.0) FiO_2

For children less than 5 years old, the oxygen regimen may need to be adjusted to a lesser concentration and flow rate, while still requiring acute level of care due to the patient's size, metabolic needs, and vital sign instability.

42:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

43:

High flow nasal cannula (HFNC) consists of the following three parameters: temperature, fraction of inspired oxygen (FiO_2), and flow rate. Each parameter is independent of each other. The flow rate is how fast the heated and humidified air/oxygen is delivered per minute and is documented as liters per minute (L/min). The flow rate is often based on the patient's weight in smaller children (Kwon, Clin Exp Pediatr 2020, 63: 3-7). To calculate the liters per kilogram per minute (L/kg/min), the flow rate is divided by the patient's weight.

Example:

HFNC flow rate: 12 L/min

Patient weight: 8 kg

$12/8 = 1.5$

This patient is on a flow rate of 1.5 L/kg/min

44:

The "R on T phenomenon" is the superimposition of an ectopic beat on the T wave of a preceding beat. Early observations suggest that R on T is likely to initiate sustained ventricular tachyarrhythmias.

45:

When T and U waves fuse, they form a TU fusion wave on an electrocardiogram. Severe hypokalemia may be associated with a large U wave created by T and U wave fusion (Zieg et al., Acta Paediatr 2016, 105: 762-72; Levis, Perm J 2012, 16: 53).

46:

The U wave occasionally follows the T wave on an electrocardiogram. A prominent upright U wave is associated with hypokalemia. When the U wave exceeds the T wave amplitude, the serum potassium level is less than 3 mEq/L. Severe hypokalemia may be associated with a giant U wave created by T and U wave fusion (Sandau et al., Circulation 2017, 136: e273-e344; Zieg et al., Acta Paediatr 2016, 105: 762-72).

47:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

48:

Prior to discharge, patients or their caregivers should demonstrate understanding of diabetes self-management (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

49:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

50:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

51:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

52:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

53:

Cardiovascular disease (CVD) is the major cause of mortality and morbidity in patients with diabetes mellitus (DM). A blood pressure level less than 130/80 mmHg in adolescents aged 13 years or older or less than the 90th percentile for age, sex, and height is recommended in individuals with diabetes to reduce CVD mortality. Patients with DM should be educated about lifestyle modifications that effectively reduce hypertension and dyslipidemia. These modifications may include (American Diabetes Association, Diabetes Care 2024, 47: S1-S314):

- Weight loss, if overweight
- Dietary Approaches to Stop Hypertension (DASH), including reducing sodium and increasing potassium intake
- Reduction of saturated fat, trans fat, and cholesterol consumption
- Increase of n-3 fatty acids, viscous fiber, and plant stanols/sterols consumption
- Moderation of alcohol consumption
- Increase in physical activity

Initial therapy for abnormal lipids should focus on controlling blood glucose levels and medical nutrition therapy that consists of limiting calories from fat to 25-30%(0.25-0.30) and saturated fat less than 7%(0.07), a cholesterol daily limit less than 200mg/day, obtaining 10%(0.10) of calories from monounsaturated fats, and avoiding trans fats. Statin therapy may be considered for youth after 10 years of age who continue to have LDL cholesterol greater than 160 mg/dL (4.1 mmol/L) or LDL cholesterol greater than 130 mg/dL (3.4 mmol/L) and one or more CVD risk factors despite lifestyle and nutrition modifications. Therapy goals aim to achieve an LDL cholesterol value less than 100 mg/dL (2.6 mmol/L) (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

54:

Large prospective randomized studies have shown that intensive diabetes management to achieve near-normal glucose levels delays the onset of nephropathy and diabetic retinopathy. Improved glycemic control has also been shown to modestly slow the progression of neuropathy. Reducing hypertension is also an effective way of preventing the progression or development of nephropathy and retinopathy (American Diabetes Association,

Diabetes Care 2024, 47: S1-S314).

55:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

56:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

57:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

58:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

59:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

60:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

61:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

62:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

63:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

64:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

65:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

66:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

67:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs

- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

68:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

69:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

70:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

71:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

72:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.